A Survey of State of the Art Methods Employed in the Offline Signature Verification Process

Michael Stauffer, Paul Maergner, Andreas Fischer, and Kaspar Riesen

Abstract Handwritten signatures are of eminent importance in many business and legal activities around the world. That is, signatures have been used as authentication and verification measure for several centuries. However, the high relevance of signatures is accompanied with a certain risk of misuse. To mitigate this risk, automatic signature verification was proposed. Given a questioned signature, signature verification systems aim to distinguish between genuine and forged signatures. In the last decades, a large number of different signature verification frameworks have been proposed. Basically, these frameworks can be divided into online and offline approaches. In the case of online signature verification, temporal information about the writing process is available, while offline signature verification is limited to spatial information only. Hence, offline signature verification is generally regarded as the more challenging task. The present chapter reviews the field of offline signature verification and presents a comprehensive overview of methods typically employed in the general process of offline signature verification.

Keywords Offline signature verification · Image preprocessing · Handwriting representation · Signature classification

M. Stauffer · K. Riesen (✉)
University of Applied Sciences and Arts Northwestern Switzerland, Institute for Information Systems, Riggenbachstrasse 16, 4600 Olten, Switzerland
e-mail: kaspar.riesen@fhnw.ch

M. Stauffer
e-mail: michael.stauffer@fhnw.ch

P. Maergner · A. Fischer
Department of Informatics, University of Fribourg, Boulevard de Pérolles 90, 1700 Fribourg, Switzerland
e-mail: paul.maergner@unifr.ch

A. Fischer
e-mail: andreas.fischer@unifr.ch

A. Fischer
University of Applied Sciences and Arts Western Switzerland, Boulevard de Pérolles 80, 1700 Fribourg, Switzerland

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1 Introduction

For several hundred years, *handwritten signatures* have been used as biometric authentication and verification measure in a broad range of business and legal transactions around the world. In fact, the first use of handwritten signatures can be traced back to the fourth century, where a signature was used to protect the *Talmud* (i.e. a central text in *Rabbinic Judaism*) against possible modifications. Later on, signatures (i.e. so-called *subscripto*) were used to authenticate documents in the Roman Empire during the period of Valentinian III [14]. More recently, several national and international digitization efforts for electronic signatures have been proposed. Yet, handwritten signatures remain very popular [8, 19, 20]. This is mostly due to the ubiquitous applicability as well as the overall high availability and acceptance of handwritten signatures when compared with electronic signatures.

Due to the high commercial and legal relevance, handwritten signatures also expose a certain risk of misuse. To mitigate this risk, *signature verification* can be employed. That is, a *questioned* signature is compared with a set of *reference* signatures in order to distinguish between *genuine* and *forged* signatures [20]. Traditionally, this task has been conducted by human experts as part of *graphology*, i.e. the study of handwriting. However, due to increasing number of documents, automatic signature verification systems have been proposed already some decades ago [28]. Actually, manual signature verification has been continuously replaced since then. Today, automatic signature verification is still regarded as a challenging pattern recognition task, and thus, remains a very active research field [8].

In general, automatic signature verification approaches can be divided into *online* (also termed *dynamic*) and *offline* (also termed *static*) approaches [19]. In the case of online signature verification, signatures are acquired by means of an electronic input device, such as a digital pen, a digital tablet, or via input on a touch screen (e.g. smartphone or handheld device). Hence, dynamic temporal information about the signing process (e.g. acceleration, speed, pressure, and pen angle such as altitude and azimuth) can be recorded during the handwriting process. In contrast, offline acquired signatures are digitized by means of scanning. As a result, the verification task is solely based on the (x, y)-positions of the handwriting (i.e. strokes), and thus, offline signature verification is generally considered the more challenging case. Note that the present chapter focuses only on offline signature verification.

Signature verification systems can also be distinguished with respect to the representation formalism that is actually used, viz. *statistical* (i.e. vectorial) and *structural* representations. The vast majority of signature verification approaches make use of statistical representations [8, 19]. That is, feature vectors or sequences of feature vectors are extracted from handwritten signature images. In early approaches, features are, for example, based on global handwriting characteristics like contour [6, 16], outline [10], projection profiles [30], or slant direction [2, 13]. However, more generic and local feature descriptors (i.e. descriptors that are not limited to handwriting images) were also used such as *Histogram of Oriented Gradients (HoG)* [41] and *Local Binary Patterns (LBP)* [12, 41]. In recent years, features have increasingly been extracted by means of a learned statistical model, viz. *Deep Learning* approaches like *Convolutional Neural Networks (CNNs)* [7, 18, 26]. Regardless of the actual type of feature, different statistical classifiers and/or matching schemes have been employed to distinguish genuine from forged signatures. Known approaches are for example *Support Vector Machines (SVMs)* [10, 12, 41], *Dynamic Time Warping (DTW)* [6, 30], and *Hidden Markov Models (HMMs)* [10, 13].

In contrast to statistical representation formalisms, structural (i.e. string-, tree-, or graph-based) approaches allow representing the inherent topological characteristics of a handwritten signature in a very natural and comprehensive way [27, 33]. Early structural approaches are, for example, based on the representation of stroke primitives [33], as well as critical contour points [4]. In a recent paper, graphs are used to represent characteristic points (so-called *keypoints*) as well as their structural relationships [27]. Hence, both the structure and the size of the graph is directly adapted to the size and complexity of the underlying signature. Moreover, these graphs are able to represent relationships that might exist between substructures of the handwriting.

However, the power and flexibility of graphs is accompanied by a general increase of the computational complexity of many mathematical procedures. The computation of a similarity or dissimilarity of graphs, for instance, is of much higher complexity than computing a vector (dis)similarity. In order to address this challenge, a number of fast matching algorithms have been proposed in the last decade that allow comparing also larger graphs and/or larger amounts of graphs within reasonable time (e.g. [15, 32]).

The goal of the present chapter is twofold. First, the chapter aims to provide a comprehensive literature review of traditional and recent research methods in the field of offline signature verification. Second, the chapter provides a profound analysis of the end-to-end process of signature verification. In particular, the chapter discusses the problems and solutions of data acquisition, preprocessing, feature extraction, and signature classification. Hence, this chapter serves as a literature survey as well as a possible starting point for researchers to build their own process pipe for signature verification.

The remainder of this chapter is organized as follows. Section 2 presents a generic signature verification process. Subsequently, the most important processing steps are individually reviewed in the following sections. First, different publicly available offline signature datasets are compared in Sect. 2.1, while common image preprocessing methods are examined in Sect. 2.2. Next, Sect. 2.3 compares different statistical and structural representations that are eventually used for classification (reviewed in Sect. 2.4). Finally, Sect. 3 presents conclusions and possible trends in the field of signature verification.

2 Signature Verification Process

Most signature verification systems are based on four subsequent processing steps, as shown in Fig. 1 [8, 20]. First, handwritten signatures are digitized by means of scanning (in Sect. 2.1 this particular step is discussed in detail). In the second step, signatures are segmented and preprocessed to reduce variations (discussed and reviewed in Sect. 2.2). Next, preprocessed signature images are used to extract statistical (i.e. vectorial) or structural (i.e. string, tree, or graph) representations (discussed and reviewed in Sect. 2.3). Finally, a questioned signature q of claimed user u is compared with all reference signatures $r \in R_u$ by means of a specific dissimilarity measure (discussed and reviewed in Sect. 2.4).

Roughly speaking, if the minimal dissimilarity to all references is below a certain threshold, the unseen signature q is regarded as genuine, otherwise q is regarded as forged. In general, the larger the reference set R, the better the accuracy of the verification process. However, the acquisition of arbitrarily large reference sets is often not possible in practice, and thus, limited to a few reference signatures per user. In the following sections, each process step is reviewed separately.

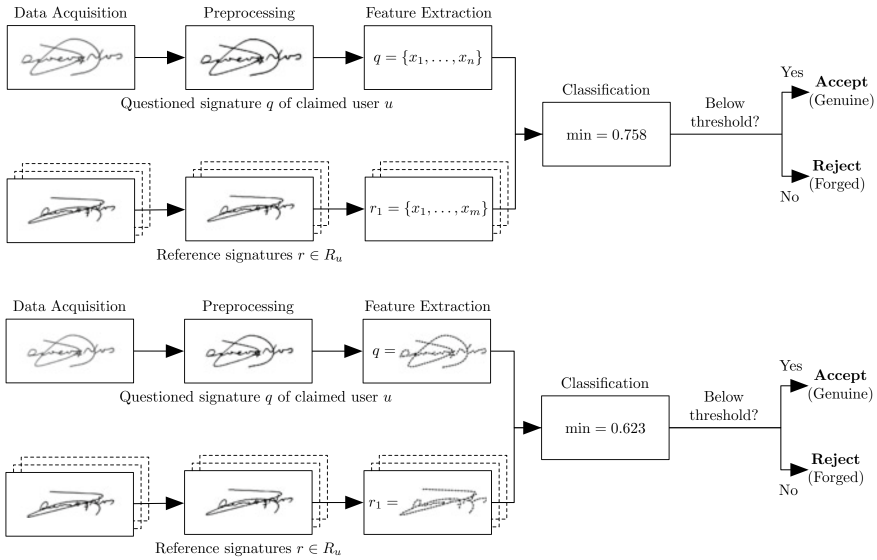


Fig. 1 Statistical (**a**) and structural (**b**) signature verification processes consist of four major steps: First, handwritten signatures are digitized by means of scanning. Second, noise and variations are reduced by means of different preprocessing algorithms. Next, preprocessed and filtered signature images are represented by either a statistical or a structural representation. In either case, a questioned signature q of claimed user u is compared with the corresponding set of reference signatures $r \in R_u$ of user u. If the minimal dissimilarity between q and $r \in R_u$ is below a certain threshold, q is regarded as genuine, otherwise q is regarded as forged

2.1 Data Acquisition and Available Datasets

During data acquisition, the available signature images are digitized into a machine-processable format by means of scanning. In most cases, digitized signatures are based on greyscale images, commonly defined as [37]

$$\{S(x, y)\}_{0 \leq x \leq X, 0 \leq y \leq Y},$$

where $S(x, y)$ denotes the greyscale level at the (x, y)-position of the image and X, Y refer to the maximum values of the x- and y-axis of the image. Typically $S(x, y)$ lies between 0 (= black) and 255 (= white).

In the last decades, a number of different private and public datasets for offline signature has been proposed, see [8] for an exhaustive overview. These datasets can be characterized with respect to their writing style (e.g. Arabic, Bengali, Chinese, Persian, Western, etc.) as well as the number of genuine and forged signatures per user. Table 1 summarizes four widely used and publicly available datasets. In particular, the table shows summaries of three datasets with real world signatures (i.e. MCYT-75 [13, 29], UTSig [34], and CEDAR [21]) as well as one synthetic dataset (i.e. GPDSsynthetic-offline [11]) for two different writing styles (i.e. Western and Persian). Note that GPDSsynthetic is replacing the widely used GPDS-960 dataset [40] that is no longer available due to new data protection regulations.

2.2 Preprocessing of Signature Images

Once handwritten signatures are digitally available, preprocessing basically aims to reduce variations caused by the image itself (e.g. noisy background, skew scanning, segmentation from the background) as well as variations caused by inter- and intrapersonal variations in the handwriting (e.g. scaling and slant of handwriting). In most cases, preprocessing is based on several subsequent steps. First, signatures are extracted from the background (e.g. bank checks). Next, a certain filtering and binarization is employed before different normalization techniques are typically applied.

Table 1 The writing style, the number of users, the number of genuine and forged signatures per user, and the resolution of the images in dpi of widely used datasets for offline signature verification

Name	Style	Users	Genuine	Forgeries	dpi
MCYT-75	Western	75	15	15	600
GPDSsynthetic-offline	Western	4000	24	30	600
UTSig	Persian	115	27	42	600
CEDAR	Western	55	24	25	300

(a) Original

(b) Segmentation (c) Filtering (d) Binarization (e) Skeletonization

Fig. 2 Common image preprocessing steps for GPDSsynthetic-Offline [11]: **a** bank check with signature,[4] **b** segmented signature image, **c** filtering by means of Difference of Gaussian, **d** binarization by means of thresholding, **e** skeletonization by means of morphological operators

Note, however, that the process sequence is not given *a priori*. The following paragraphs give an overview of some common methods and techniques that are widely accepted as standard preprocessing steps. These steps can be divided into four main phases (see also Fig. 2):

1. Segmentation
2. Filtering
3. Binarization
4. Skeletonization

For an exhaustive overview on preprocessing techniques, see [8, 20].

1. Segmentation

As illustrated in Fig. 2a, the signatures are in many cases part of some larger document (e.g. bank checks or forms), and thus, need to be extracted first. Especially bank checks are regarded as a challenging segmentation problem due to colored background images, logos and preprinted lines. To address this issue, different segmentation approaches have been proposed in the literature [2, 12, 13, 33]. In [9], for example, the segmentation is based on subtracting a blank check from the signed artifact. More generic approaches, are based on *Speeded Up Robust Features (SURF)* features in combination with an Euclidean distance metric [1]. Recently, different Deep Learning techniques based on CNNs have been proposed for this task [22, 34]. In general, the importance of signature segmentation is decreasing due to two reasons: First, the increasing use of blank signature fields and second, the declining importance of bank checks.

2. Filtering

In many cases, scanned handwritten signatures are affected by noisy background, i.e. small pixel fragments that do not belong to the actual signature. The reasons for that can be filthy scanners and papers, smearing as well as non-smooth writing surfaces, to mention just a few. To reduce noise, different filter methods are often employed such as *low pass* filter [4, 12, 24], *median* filter [3], *morphological* filter [2, 6, 13, 16] as well as *Difference of Gaussian* filter [27].

3. Binarization

To clearly distinguish between foreground and background pixels in an image, scanned signature artifacts are often binarized [6, 27]. That is, each pixel of a greyscale image is either assigned black (i.e. ink of signature) or white (i.e. background). In many cases, binarization leads to better classification results as well as overall lower processing times. A number of different binarization approaches have been proposed for this particular task such as *Niblack's* algorithm [4], *Otsu's* method [12, 16–18, 23, 24] as well as *thresholding* [2, 4, 13, 27, 35].

4. Skeletonization

Further image preprocessing approaches are mainly aimed at the reduction of intrapersonal variations in the handwriting. In [2, 3, 13], for example, the handwriting scaling is corrected, while skew and slant are corrected in [3, 23, 24, 30]. Intrapersonal variations can also be reduced by means of skeletonization that is also known as thinning [3, 4, 23, 24, 27].

Note that a number of Deep Learning approaches for signature verification have been proposed recently [8]. In these cases, the neural network is often restricted to a fixed size input, and thus, a certain resizing is necessary [7, 18]. Moreover, the size of training data is often rather small in the case of signature verification which might be a problem in the paradigm of Deep Learning. Hence, data augmentation is often employed in this case to artificially increase the number of available training data [17, 18].

2.3 Feature Extraction

Based on segmented and preprocessed signature images, different types of statistical and structural representations can be extracted. The following two subsections give a brief overview of both representation types.

2.3.1 Statistical Representations of Signatures

Statistical representations are extracted from handwritten signature images either on a *global* or *local* scale [37]. In the former case (also known as *holistic approach*), features are extracted on complete signature images, and thus, represented by a single (typically high-dimensional) feature vector $\mathbf{x} \in \mathbb{R}^n$. In the case of local representations, subparts of the handwritten signature images are independently represented by local feature vectors. For example, features are extracted by means of a sliding window that moves from left to right over the whole signature image. Then, at each window position a set of different features is extracted. As a result, a linear sequence of m local feature vectors is acquired for every signature image, i.e. $\mathbf{x}_1, \ldots, \mathbf{x}_m$ with $\mathbf{x}_i \in \mathbb{R}^n$.

In early approaches, features are often used to represent different global handwriting characteristics [37]. For instance, features that are based on the direction of the contour, as shown in Fig. 3a [6, 16]. Similarly, the handwriting outline has also been used as representation formalism in [10]. Moreover, the direction of slant, i.e. the inclination of the handwriting, has been found to be useful [2, 13]. Last but not least, projection profiles, i.e. the number of foreground pixels in the horizontal or vertical direction, have been employed as global features in [30].

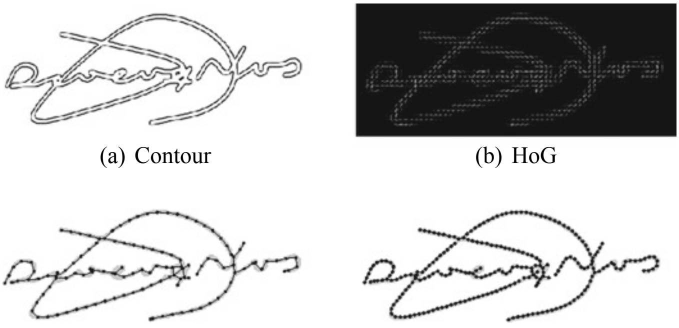
In contrast, features can also be extracted from a local perspective on the signature [37]. For example, different morphological features have been proposed in [23]. In case of *Gradient, Structural and Concavity (GSC)* features, three types of features are extracted by means of a grid-wise segmentation and concatenated to form a feature vector [35]. Global features (e.g. the signature height, projection profiles, slant angle, etc.) can also be combined with different local features (e.g. grid and texture features), as shown in [3].

Further local features are often based on more generic texture descriptors, i.e. features that can be used to represent arbitrary patterns [37]. In [41], for example, HoG have been employed, as illustrated in Fig. 3b. That is, the direction of gradients is first extracted at different local cell positions and then concatenated to form a single histogram. Likewise, LBP are used to describe circular structures of an image at different neighborhood levels [12, 41], while so-called *surroundedness* features are extracted in [24].

Finally, it is worth mentioning that a number of Deep Learning approaches for feature extraction have been proposed in the last years [18, 26, 36]. That is, rather than extracting handcrafted features by means of a certain predefined procedure, this type of feature representation is learned in an unsupervised manner by means of a CNN. In particular, different CNN architectures have been proposed for this task such as *spatial pyramid pooling* [17], *triplet network* [26] as well as *siamese network* [7].

2.3.2 Structural Representations of Signatures

Structural representations are based on powerful data structures such as strings, trees, or graphs [4, 27, 33, 37]. In general, a graph is defined as a four-tuple $g = (V, E, \mu, \nu)$ where V and E are finite sets of nodes and edges, $\mu : V \rightarrow L_V$ and $\nu : E \rightarrow L_E$ are labeling functions for nodes and edges, respectively [31]. In general, the graph size is not fixed and thus graphs are—in contrast to global fea-



(a) Contour (b) HoG

(c) Keypoint Graph (d) Inkball Model

Fig. 3 Visualization of contour features (see Fig. 3a), HoG features (see Fig. 3b), keypoint graph (see Fig. 3c), and inkball model (see Fig. 3d) of one signature from GPDSsynthetic-Offline [11]

ture vectors—flexible enough to adapt their size to the size and complexity of the underlying handwritten signature image. Moreover by means of edges, graphs are capable of representing binary relationships that might exist in different subparts of the underlying signature.

A vast amount of signature verification approaches still makes use of statistical rather than structural approaches [8, 19, 37]. This is rather surprising as the inherent representational properties of structural formalisms would be well-suited to represent handwriting.

An early structural signature verification approach is based on the representation of signatures by means of stroke primitives [33]. Another approach is based on the detection of characteristics points—so-called keypoints—on contour images [4]. Following this line of research, a recent approach is based on the detection of keypoints in skeletonized signature images, as illustrated in Fig. 3c [27, 37]. In this particular scenario, nodes are used to represent keypoints (e.g. start, end, and intersection points of the handwritten strokes), while edges are used to represent strokes between these keypoints. Other graph-like representations, that have become more popular in recent years are, for example, *inkball* models, as illustrated in Fig. 3d [25]. Similar to keypoint graphs, nodes in inkball models are based on characteristic points of the skeleton (i.e. junction and end points). Note, however, that inkball models are in contrast with keypoint graphs based on rooted trees rather than general graphs.

2.4 *Classification of Signatures*

Finally (and regardless the actual representation formalism used), the signature verification is based on a classification task. That is, a questioned signature q of claimed user u is compared with all reference signatures $r \in R_u$ of registered user u (see

Fig. 1). However, the actual method that determines whether q is regarded as genuine or forged depends on the chosen representation formalism, i.e. whether signatures are represented by means of a statistical or structural formalism. The following two subsections give an overview of both approaches.

2.4.1 Statistical Approaches for Signature Verification

In case of sequences of feature vectors (i.e. the signature is represented by $\mathbf{x}_1, \ldots, \mathbf{x}_m$), the classification task is often based on a dynamic programming approaches such as, for example, DTW [6, 30]. That is, two sequences of feature vectors (i.e. one sequence for q and one sequence for r) are optimally aligned along one common time axis. The minimal sum of alignment costs can then be used as dissimilarity measure between q and r. In many cases, however, signatures are represented by holistic and often high-dimensional feature vectors (i.e. the signature is represented by $\mathbf{x} \in \mathbb{R}^n$). Consequently, feature vectors are compared by means of different distance measures like for example Euclidean distance [7, 10, 26], χ^2 distance [12, 16], or Mahalanobis distance [2, 13].

In contrast to these so-called *learning-free* methods, a number of *learning-based* approaches have been proposed as well [12, 13, 17]. That is, a statistical model is trained *a priori* on a (relatively large) training set of reference signatures. Different statistical models have been employed for this task for example SVMs [10, 12, 17, 18, 23, 24, 41], HMMs [10, 13] as well as Neural Networks [3, 23, 24].

Generally, learning-based approaches lead to lower error rates when compared with learning-free methods. However, this advantage is accompanied by a loss of flexibility and generalizability, which is due to the need to learn a statistical model on some training data. In fact, the accuracy of many learning-based approaches is crucially depending on the size and quality of the labelled training data. However, in case of signature verification the acquisition of training data is often expensive and/or otherwise limited.

2.4.2 Structural Approaches for Signature Verification

Structural representations, in particular graph-based representations, offer some inherent representational advantages as discussed before. However, this advantage is accompanied with an increased complexity with respect to basic dissimilarity measures. In fact, the exact computation of a graph dissimilarity measure on general graphs is known to be $\mathcal{NP}$-complete, and thus several fast approximations for this task have been proposed in the last decades [31]. In an early work, stroke primitives are first merged locally and then compared by means of a global dissimilarity measure [33]. Moreover, keypoints of contour images are matched by means of a point matching algorithm in [4]. However, the actual structural relationships of the handwriting are not considered during these first structural matching procedures.

More recent structural signature verification approaches make use of fast approximations for *Graph Edit Distance (GED)* [15, 32]. GED measures the minimum amount of distortion (i.e. insertion, deletion, substitution of both nodes and edges) needed to transform one graph into another. In the case of tree-based inkball models, the classification is based on an energy function that measure the amount of deformation, similar to GED [25].

Recently, different approaches have been proposed that allow the combination of statistical and structural classification approaches [26, 38, 39]. In [38], for example, subgraphs are first matched by means of graph matching and then optimally aligned by means of DTW. Another approach is based on an *ensemble* method that allows the combination of a Deep Learning approach (i.e. triplet network) with a fast GED approximation [26]. Finally, *prototype* graph embedding allows to embed graphs into a vectorial representation [39]. This approach combines the representational power of graphs and the mathematically well-founded vector space for dissimilarity computation.

3 Conclusion and Outlook

Handwritten signatures have been an important verification measure for many business and legal transactions for several centuries [37]. The popularity of handwritten signatures remains high, even though several international initiatives for electronic signature have been announced. That is, handwritten signatures offer some inherent advantages such as ubiquitous applicability as well as the overall high acceptance and availability. However, handwritten signatures expose also a certain risk of misuse. For this reason, automatic signature verification has been proposed [8, 19, 20]. Most signature verification frameworks are based on four processing steps. First, signatures are digitized by scanning. Second, scanned handwritten signature images are typically preprocessed in order to reduce variations caused by both the artifact (e.g. noisy background, skew scanning) and the handwriting (e.g. scaling of the handwriting). Next, preprocessed signature images are represented either by a statistical (i.e. vectorial) or structural (e.g. graph-based) formalism. Finally, a questioned signature is compared with a set of reference signatures in order to distinguish between genuine and forged signature.

The present chapter provides a comprehensive overview of the state of the art in signature verification based on offline data. This chapter reviews in particular traditional and very recent methods used in the various steps in a generic process of offline signature verification. This analysis can be used either as a literature review or as a starting point for developing an own signature verification engine.

In general, structural approaches offer some inherent representational advantages when compared with statistical formalisms. That is, graphs are able to adapt both their size and structure to the underlying pattern. Moreover, graphs are able to represent a binary relationship that could exist between subparts of the handwriting. However, most signature verification approaches still make use of a statistical rather than structural representation. This is mainly due to the fact that algorithms and methods that need graphs as input have a substantially greater complexity than their statistical counterparts. Yet, in recent years, several powerful and efficient methods for graph processing have been proposed [31]. These methods enable the use of graphs in the field of signature verification. In the future, great potential for further structural signature verification approaches can be expected, especially with the rise of Deep Learning methods for graphs, the so-called Geometric Deep Learning [5].

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